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2048 Pixel Shortwave Infrared InGaAs Focal Plane Arrays for HY3A Ocean Satellite Multispectral Imaging Applications

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***Abstract*—We report on the design and development of $In_{0.53}Ga_{0.47}As/InP$ and $In_{0.75}Ga_{0.25}As/In_{0.52}Al_{0.48}As$ shortwave infrared focal plane array (FPAs) in formats of 2048×52 and 2048×12, respectively, deployed in the multispectral imaging instrument on the HY3A ocean satellite. The array length of the extended-wavelength $In_{0.75}Ga_{0.25}As$ FPA is as far as know the longest for aerospace applications. Four center wavelength optical filters of 1.020, 1.245, 1.640 and 2.135 μm are integrated on the FPA modules densely, forming of two space-qualified FPA detector assemblies. Dedicated low noise readout integrated circuits by incorporating of a triple sampling and hold architecture are designed. Low dark current densities of 48 and 320 nA/cm^2, high quantum efficiencies of >80%, low dark signals of 20 mV, low readout noises of 1.8 and 1.1 mV, and good response nonuniformities of 3.0% are simultaneously obtained for all spectral channels for the two FPA assemblies under chip temperatures of 293 and 223 K, respectively. High pixel operabilities of 99.99% and 99.78% are achieved for the two FPAs with peak detectives of 4.0×10^{12} and 1.4×10^{12} cmHz$^{1/2}$/W. Modulation transfer functions of higher than 0.5 are measured for all spectral channels. Laboratory scanning imaging are demonstrated for different channels, showing very high spatial resolution, contrast and response sensitivities.**

***Index Terms*—Focal plane array, SWIR, InGaAs, extended-wavelength, ocean satellite, detector assembly.**

I. INTRODUCTION

HIGH-resolution multi- and hyper-spectral imaging in the 1-2.5 μm shortwave infrared (SWIR) band plays the most important role in the satellite-based remote sensing for earth resources, such as land, ocean, cloud, mineral and etc. The high performance HgCdTe [1, 2] and $In_xGa_{1-x}As$ ($0.53\leqslant x<1$) [3, 4] focal plane arrays (FPAs) in the SWIR band are the two primary detector candidates for the aerospace sensing applications. Benefiting from the relatively high operating temperature, good material uniformity, and lower dark current, both the lattice-matched $In_{0.53}Ga_{0.47}As/InP$ and the extended-wavelength $In_{0.60}Ga_{0.40}As/InP$, $In_{0.81}Ga_{0.19}As/InP$, and $In_{0.83}Ga_{0.17}As/InP$ linear detector arrays with pixels of up to 1024 are incorporated in the near-infrared (NIR) channels of the scanning imaging absorption spectrometer for atmospheric chartography (SCIAMACHY) instrument in the ENVISAT satellite in 2002 [5], which is also the first use of the InGaAs FPAs in space remote sensing applications. While the dark current densities (J_d) of 0.14 nA/cm^2 at 200 K, 0.15 nA/cm^2 at 200 K, 1.8 nA/cm^2 at 147 K, and 1.2 nA/cm^2 at 146 K were achieved for these FPAs, the lattice-mismatched $In_xGa_{1-x}As$ FPAs saw a faster increase of J_d as well as dead pixel numbers with the increase of the cumulative dose of proton radiation exposure in comparison with the lattice-matched $In_{0.53}Ga_{0.47}As$ [6]. A larger $In_{0.53}Ga_{0.47}As/InP$ 3000×1 mosaic linear array composed of three 1024×1 linear array chips in a pixel pitch of 25 μm was deployed in the multispectral instrument of the Proba-V satellite in 2010 for spectral imaging in a broad band of 0.9-1.7 μm [7]. Furthermore, two linear arrays with a shrank and extended cutoff wavelengths of 1.55 and 2.21 μm, respectively, were equipped in the spectral imager in the EarthCare satellite [8] for earth ground mapping in 2014. In regards to the SWIR InGaAs FPAs, the key challenges rely on the mitigation of the surface-related leakage current in the planar and mesa photodiodes, the suppression of the dislocation-related tunneling dark current in the extended-wavelength materials, and the improvement of the response nonuniformity over the whole pixel array.

Due to the extremely faint signal level in the space remote SWIR sensing scenarios, especially for spectral imaging with filtered narrow band down to several nanometers, relatively long integration times (T_{int}) and high operational amplifier gains are generally required, which put forward higher performance standards on the dark current, the dark noise and the associated pixel nonuniformities of the SWIR FPA detectors. For ocean color satellites that are primarily designed for sensing of the radiometric information of phytoplankton in

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marine biogeochemistry, the global carbon cycle and the response of marine ecosystems, the frequently used spectral bands are 400-12000 nm [9] among which the 900-2200 nm SWIR bands are especially important for near shore imaging and atmospheric calibration owing to the strong absorption of CO_2 and H_2O molecules [10]. The SWIR HgCdTe FPA sensors have been serving as the workhorse for ocean color satellites instruments. However, the low operating temperatures down to 110 K, high power consumption of the cooler machine, and heavy detector assemblies are concomitant requirements for the system design. With the progressive progresses on the control of the material defects as well as the improved device processing technology, high performance $In_xGa_{1-x}As$ ($0.53 \leq x < 1$) FPA detectors that cover the same SWIR band are also available for ocean color satellites instruments. Both the lattice-mismatched $In_{0.53}Ga_{0.47}As$ and the HgCdTe FPA detector assemblies with a SWIR spectral range of 940-2260 nm are integrated in the Ocean Color Instrument (OCI) on the Plankton, Aerosol, Cloud, ocean Ecosystem (PACE) satellite mission [11]. The programmable mid-resolution spectral imager (PMRSI) payload in the latest generation ocean satellite of HY3A in China is designed for observing of the land, ocean, silt, and water pollution with a swath width of 1000 km and a ground resolution of 100 m, which had been already launched in 16 Nov. 2023. Five imaging cameras were included in this instrument among which each camera was equipped two SWIR FPA assemblies, covering the bands of 1000-2250nm. InGaAs FPA detector assemblies with a large array length of up to 2048 were designed and built. The high signal to noise ratio (SNR) requirement of the mission has posed very high challenge on the performances of the SWIR FPA detectors, especially for the extended-wavelength FPA chips. To satisfied the performance requirements, advances on the dielectric surface passivation, material defect suppression, as well as stray light suppression are in demand.

In this paper, the design and the performance characterization of the 2048×52 $In_{0.53}Ga_{0.47}As$/InP and the 2048×12 $In_{0.75}Ga_{0.25}As$/$In_{0.52}Al_{0.48}As$ SWIR FPA assemblies deployed in the multispectral imaging instrument of the HY3A satellite are reported in detail. The photodiode epitaxial structures and processing technologies, the spectral channel structures, the readout integrated circuit (ROIC) architectures, and the assembly structures are discussed. Extensive performance measurements on the dark current, response spectra, light response signal, dark signal, dark noise, modulation transfer function (MTF) are also reported. Laboratory scanning imaging demonstrations showing fine spatial resolution and good response sensitivities.

II. FPA DETECTOR ASSEMBLY DESIGN

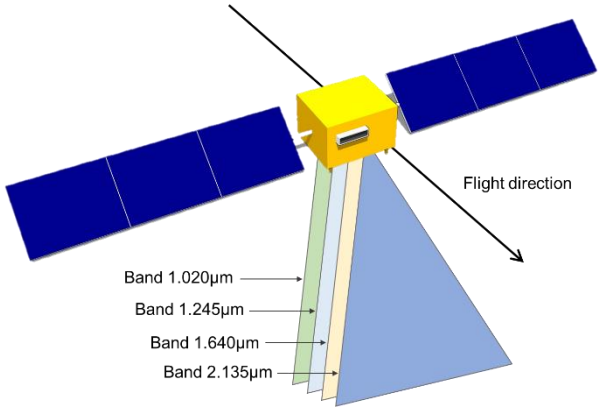


Fig. 1. A schematic illustration for the operation of the push-broom imaging of the earth by the mid-resolution spectral imager payload on the HY3A satellite. Four SWIR channels with center wavelengths of 1.020, 1.245, 1.640 and 2.135 μm are integrated with this imager.

A. Butting FPA module design

The PMRSI payload in the HY3A satellite was operated in a push-broom imaging mode of the earth on the track along the flight direction, as shown in Fig.1. Four parallel SWIR imaging channels with center wavelengths of 1020, 1245, 1640 and 2135 nm, respectively, were designed on this instrument. While the 1020, 1245 and 1640 nm channels were integrated together on a detector assembly that was built with the lattice-matched $In_{0.53}Ga_{0.47}As$ material, the 2135 nm channel was integrated on a separate detector assembly that fabricated based on the lattice-mismatched extended wavelength $In_{0.75}Ga_{0.25}As$ FPA chips. With an F/3 optics, the PMRSI covers a ground swath width of 1000 km with a resolution of 100 m. The total detector pixels required were 10000 for each SWIR channel. Five SWIR cameras were built on the PMRSI to enable a large field of view (FOV) after stitching. Each camera contains a SWIR FPA detector assembly with a pixel number of no less than 2000 perpendicular to the flight direction to ensure the overall coverage of the FOV.

Figures 2(a) and 2(b) show a three-dimensional (3D) schematic illustration of the design and a photograph for the 1.0/1.2/1.6 μm channels detector assembly (referred to as assembly A hereafter), respectively. Four butting 552×52 $In_{0.53}Ga_{0.47}As$ FPA modules with a pixel pitch of 30 μm arranged in an interleaved pattern are sequentially mounted on a ceramic interposer, forming a total array length of 2048 perpendicular to the flight direction. Some pixels are overlapped for two adjacent FPA to ensure a seamless contiguity of the FOV. Benefiting from the extremely low

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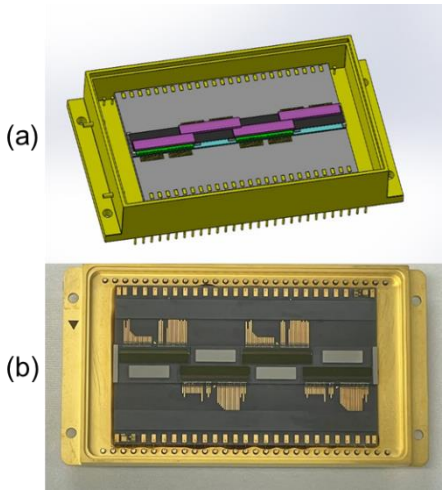


Fig. 2. (a) A 3D schematic illustration of the design and (b) a photograph of the detector assembly A. The four FPA modules are sequentially mounted on a ceramic interposer in an interleaved pattern.

dark current nature of the lattice-matched $In_{0.53}Ga_{0.47}As$ FPA chips, the in orbit operating temperature for these detector chips were designed to be 293 K, which is controlled via the thermal management from the mainframe of the instrument.

In regards to the 2.1 μm channel, four butting $In_{0.75}Ga_{0.25}As$ FPA modules with a format of 552×12 and the same pixel pitch are arranged in a similar manner to form the detector assembly (referred to as assembly B hereafter) (Fig.3). As a result of the increased dark current of the lattice-mismatched materials, these extended-wavelength FPA detector chips were designed to be operated at 223 K in orbit to maintain a high SNR. The temperature was controlled by a thermal electric cooler (TEC) and a temperature sensor in a close-loop feedback. Heat generated from the TEC was taken away via a heat tube controlled by the thermal management system.

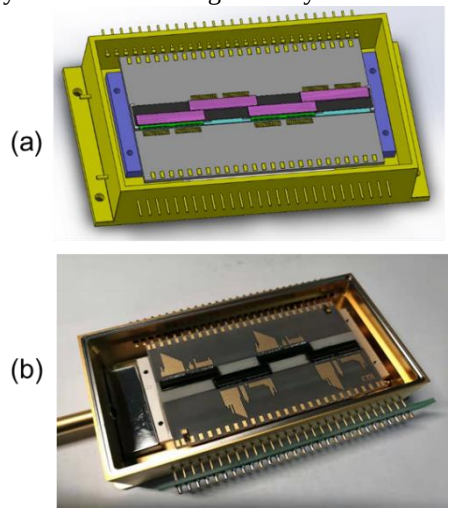


Fig. 3. (a) A 3D schematic illustration of the design and (b) a photograph of detector assembly B. The four FPA modules are also sequentially mounted on a ceramic interposer in an interleaved pattern.

B. Detector chip design

Figures 4(a) and 4(b) show the normalized spectral response curves for detector assemblies A and B, respectively. The measured temperatures were 293 and 223 K, respectively. The response wavelength ranges were 950-1650 and 1050-2250 nm for detector assemblies A and B, respectively. The narrow spectral imaging channels are also indicated. The $In_{0.53}Ga_{0.47}As$/InP FPA chips were processed based on a zinc diffused planar photodiode structure as shown in Fig. 4(c), whereas the $In_{0.75}Ga_{0.25}As$ FPA chips were fabricated based on a micro-mesa photodiode structure as shown in Fig. 4(d). For the lattice-mismatched material system, $In_{0.75}Ga_{0.25}As/In_{0.75}Al_{0.25}As$ structures were heterogeneously grown on the InP substrate with a linearly-graded InAlAs buffer layer [12, 13]. In-situ thermal treatments [14] were introduced to minimize the dislocation defect density in the $In_{0.75}Ga_{0.25}As$ absorber region in order to suppress the bulk dark current. Detail growth parameters, material characterizations and FPA fabrication recipes were described elsewhere [15, 16]. The finished $In_{0.53}Ga_{0.47}As$ and $In_{0.75}Ga_{0.25}As$ FPA chips were flip-chip bonded to self-designed 552×52 and 552×12 1silicon capacitive trans-impedance amplifier (CTIA) ROICs to form hybrid SWIR FPA modules.

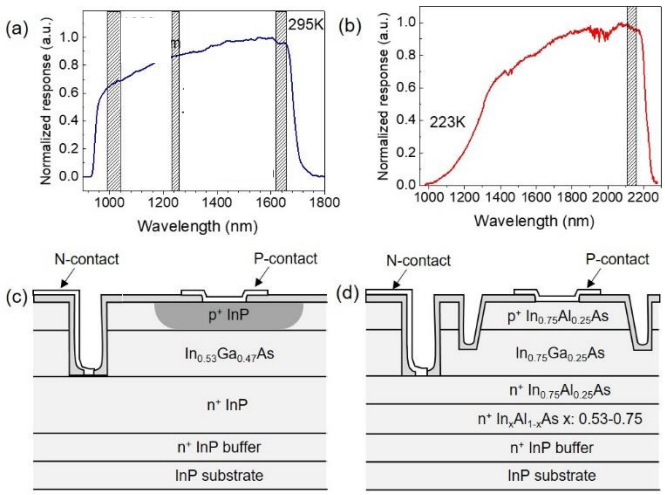


Fig. 4. (a) and (b) Spectral response curves of detector assemblies A and B, respectively. The narrow spectral imaging channels are also indicated. (c) and (d) The schematic structures of the $In_{0.53}Ga_{0.47}As$ and $In_{0.75}Ga_{0.25}As$ FPA chips for detector assemblies A and B, respectively.

C. Spectral channel design

Figures 5(a) and 5(b) show schematic illustrations for the arrangements of the narrow optical filters that were integrated on the four interleaved FPA modules for the 1.0, 1.2 and 1.6 μm channels on assembly A and the 2.1 μm channel on assembly B, respectively. Black regions represent that these pixels were masked by an aperture that was mounted above the filters. The masked pixels serve as reference pixels for the dark signal subtraction during in orbit imaging operation. Other pixels were exposed through the aperture openings, serving as active

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imaging pixels. Each SWIR channel used four consecutive pixel rows. This means the total pixel array size for all spectral channels are 2048×4. The PMRSI operated in a time-delay-integration (TDI) mode on track to enhance the SNR of image. Finally, a total pixel length of 2048 perpendicular to the flight direction after considering of the pixel overlapping.

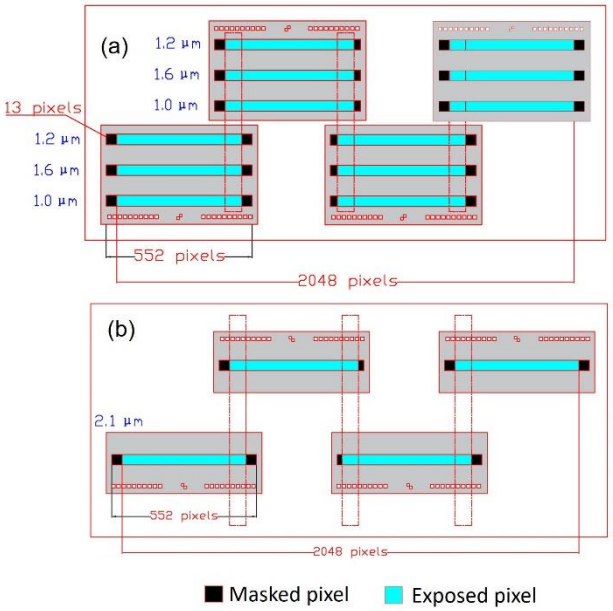


Fig. 5. Schematic illustration of the arrangements of the pixel overlap and the narrow optical filters on the four interleaved FPA modules for the (a) 1.0, 1.2 and 1.6 μm channels on assembly A and the (b) 2.1 μm channel on assembly B.

D. Low noise ROIC design

The fundamental function of the ROIC is to provide stable bias for the photodetector array, convert the weak photocurrent generated by the photodetectors into a voltage, and serially output the synchronously sampled voltage signals to the exterior of the module. The pixel circuit comprised an input stage, a sampling circuit, and a pixel output stage, as shown in Fig. 6. Considering factors such as charge injection efficiency, bias stability, and bandwidth, a CTIA structure with capacitive feedback was selected for the input stage. Correlated double sampling (CDS) was implemented under the integration-while-read (IWR) mode. The pixel output stage employed an NMOS source follower structure. In the column circuit, a low-power single-gain buffer operational amplifier was designed. The output stage utilized a push-pull output amplifier. Two ROICs with array formats of 552×52 and 552×12 were designed for the SWIR detector assemblies A and B, respectively. The pixel pitch was designed to be 30 μm, matching the pixel pitch of the InGaAs photodetector array chips. Both ROICs shared an identical architecture and modules, differing only in pixel array format.

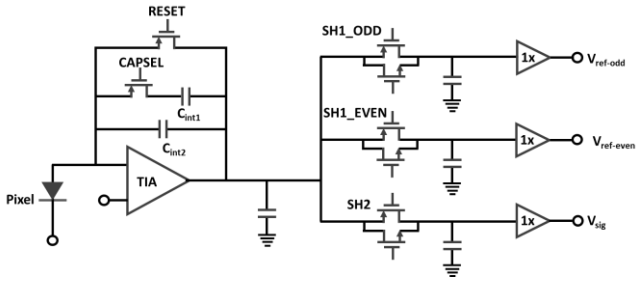


Fig. 6. Schematic diagram of the ROIC architecture for the SWIR detector assemblies A and B.

To reduce readout noise, a triple-branch sampling and hold (S/H) architecture with an intra-pixel coherent multi-sampling mis-frame interleaved readout was adopted. Specifically, three S/H branches are designed within each pixel: two branches (for odd and even frames) share a common reference (R) branch, and one branch is for the signal (S). This method of distinguishing between odd and even frames for sampling helps to mitigate the non-coherent readout noise of the ROIC [17], thereby minimizing the total dark noise of the detector. The default operating mode of the ROIC is the integration-while-read (IWR) mode. The integration time is adjustable within the range of 1 to 14 ms. Each ROIC features a single output channel (including R output and S output), with a pixel readout rate capable of reaching 10 MHz. Both ROICs are fabricated by using a 0.18 μm mixed-signal CMOS foundry process.

III. FPA PERFORMANCE TEST RESULTS

The electro-optical performances of the two 2048 InGaAs FPA assemblies including the temperature-dependent dark current of pixels, the narrow band spectral response, the light and dark signal responses, the detector readout noises, and the modulation transfer function were comprehensively characterized and discussed.

A. Dark current of FPA Chips

Figures 7(a) and 7(b) show the measured temperature-dependent dark current density -voltage (J_d-V) curves for the $In_{0.53}Ga_{0.47}As$ and the $In_{0.75}Ga_{0.25}As$ FPAs, respectively, which were directly probed on a single pixel of the FPA chips in a liquid nitrogen cooled probe station. A temperature-synchronized infrared radiation-shield was utilized to eliminate the background radiation. The probed J_d at -10 mV were 10.3 nA/cm^2 at 293 K and 175 nA/cm^2 at 223 K for the $In_{0.53}Ga_{0.47}As$ and the $In_{0.75}Ga_{0.25}As$ FPAs, respectively. The corresponding R_0A were 5.29×10^8 and 2.64×10^5 Ω·cm^2, respectively. Figure 7(c) further plots the Arrhenius relationships for the -10 mV J_d as a function of the inverse temperature. Linear dependences with different slopes for T>273 K and T>223 K were observed for both FPAs, respectively, which are modeled by using the expression of $J_d \propto \exp(-E_a/k_BT)$, where k_B is the Boltzmann constant, E_a is the activation energy [18]. The extracted E_a are 0.614 and 0.406 eV for the $In_{0.53}Ga_{0.47}As$ and the $In_{0.75}Ga_{0.25}As$ FPAs,

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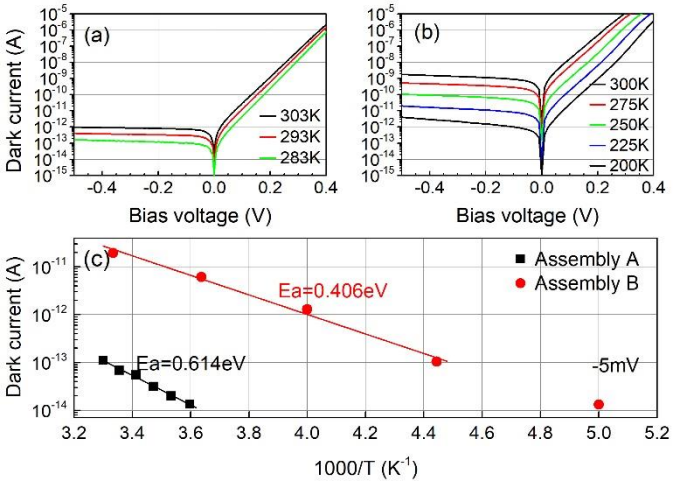


Fig. 7. (a) and (b) Measured temperature-dependent J_d-V of the $In_{0.53}Ga_{0.47}As$/InP and $In_{0.75}Ga_{0.25}As/In_{0.52}Al_{0.48}As$ FPA pixels, respectively. (c) The extracted Arrhenius plot for the pixel dark currents for the two FPAs. The fitted activation energies (E_a) are also given.

respectively, as also indicated in Fig. 7(c). Both E_a roughly agree with three quarters of the band gap of each FPA which indicates a diffusion current dominant case was achieved. In addition, the Shockley-Read-Hall (SRH) and the trap-assisted-tunneling (TAT) currents also make non-negligible contributions to the total dark current at the small bias voltages.

B. Light and dark signal responses

Characterizations of the FPAs were carried out in an in-house built measurement system with full functionalities of black-body radiation illumination, power supply, digital timing, signal amplification, acquisition, and data processing [16]. The FPA assemblies were mounted on a liquid-cooled platform with a close-loop controlled temperature. Measurements were carried out in the integration while readout (IWR) mode under a readout data rate of 1 MHz. Figs. 8(a)-8(c) plot the measured narrow band spectral response signal voltage curves of a row of pixels (2048 pixels in total) for the 1.0, 1.2 and 1.6 μm channels of detector assembly A at a T_{int}=32 ms under high-gain and at 293 K. Fig. 8(d) plots the measured narrow band spectral response signal voltage for the 2.1 μm channel of assembly B at a T_{int}=14 ms under low-gain and at 223 K. All the FPA measurements were performed under zero bias unless otherwise noted. The averaged V_s over the whole array, $\overline{V_s}$, was calculated by using the expression of

$$\overline{V_s} = \frac{1}{M \times N - (d+h)} \sum_{i=1}^{M} \sum_{j=1}^{N} V_s(i,j)$$

where M and N are the row and column numbers of the FPA, d and h are the number of dead and hot pixels, respectively, $V_s(i, j)$ is the signal of the pixel (i, j), yielding $\overline{V_s}$ of 1.532, 0.579, 0.912 and 0.155 V for the 1.0, 1.2, 1.6 and 2.1 μm channels, respectively. Meanhwile, the non-uniformities of the signal (NU_s), for both FPAs were also extracted according to the

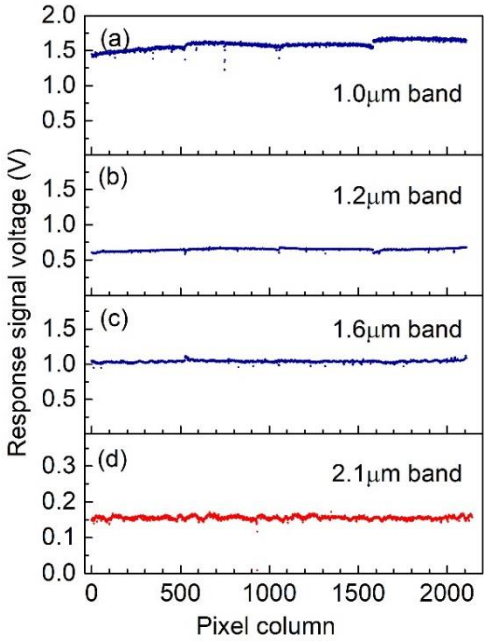


Fig. 8. Measured narrow band spectral response signal voltage curves of a row of pixels (2048 pixels in total) for FPA assemblies A at 300 K and high gain and B at 223 K low gain, respectively, under a blackbody radiation: (a) 1.0 μm band, (b) 1.2 μm band, (c) 1.6 μm band, and (d) 2.1 μm band.

following equation:

$$NU_s = \frac{1}{\overline{V_s}} \sqrt{\frac{1}{M \times N - (d+h)} \sum_{i=1}^{M} \sum_{j=1}^{N} [V_s(i,j) - \overline{V_s}]^2} \times 100\%$$

The calculated NU_s are 3.0%, 3.0%, 1.5% and 3.2% for the 1.0, 1.2, 1.6 and 2.1 μm channels, respectively. Note that these relatively low NU_s are raw parameters without correction and will be further improved on the instrument after NU correction.

Statistical histograms revealed Gaussian distributions for the four channels, as shown in Figs. 9. The slightly asymmetric distributions for the 1.0 and 1.2 μm channels are accounted by the variation of the cutoff wavelengths of the narrow band

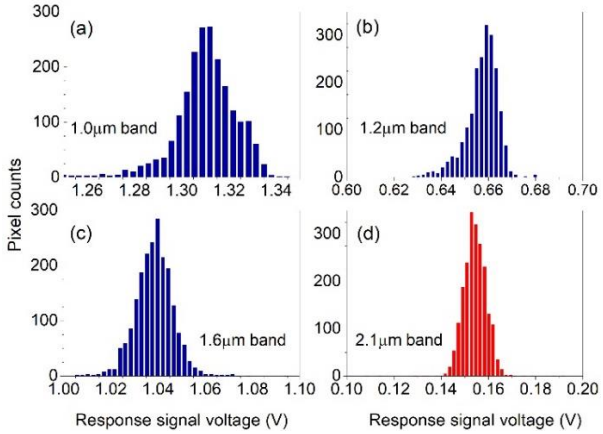


Fig. 9. Statistical histogram distributions of the measured narrow band spectral response signal voltages as shown in Fig. 8.

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optical filters. This can be further improved by optimizing the coating processes.

Figure 10(a) plots the measured dark signal voltage (V_{ds}) of pixels for the 1.0/1.2/1.6 μm FPA assembly A at a T_{int} of 32 ms and 293 K under high-gain. For the 2.1 μm FPA assembly B, the V_{ds} of pixels were measured at a T_{int} of 14ms and 223 K under low-gain, as shown in Fig. 10(b).

The calculation method is the same as that for the V_s. The averaged V_{ds} over the whole array, $\overline{V_{ds}}$, was calculated by using the expression of

$$\overline{V_{ds}} = \frac{1}{M \times N - (d+h)} \sum_{i=1}^{M} \sum_{j=1}^{N} V_{ds}(i,j)$$

where M and N are the row and column numbers of the FPA, d and h are the number of dead and hot pixels, respectively, $V_{ds(i,\,j)}$ is the dark signal of the pixel (i, j), yielding $\overline{V_{ds}}$ of 16.8 and 22.6 mV for FPA assemblies A and B, respectively. The T_{int}-dependent $\overline{V_{ds}}$ for both FPAs were further measured over a range of 0.1 to 100 ms. Linear dependences with different slopes are observed, which are fitted by using the equation of

$$\overline{V_{ds}} = \frac{\overline{J_d} \cdot A \cdot T_{int}}{C_{int}}$$

Where A is the area of a pixel, $\overline{J_d}$ is the averaged J_d over the whole array, yielding $\overline{J_d}$ of 48 nA/cm^2 at 293 K for assembly A and 320 nA/cm^2 at 223 K for assembly B, respectively. Statistical histograms of the V_{ds} of pixels revealed good Gaussian distributions for the two FPA assemblies, as shown in Figs. 11(a) and 11(b), respectively. Noteworthy is that the $\overline{J_d}$ extracted from the statistical V_{ds} are slightly higher than the J_d obtained in the low-temperature probing test (Fig.5). Such a deviation is elucidated as follows. First of all, the V_{ds} of the pixels of the $In_{0.53}Ga_{0.47}As$ and $In_{0.75}Ga_{0.25}As$ FPAs scattered from -89.4 to 33.8 mV and from -178.8 to 90.2 mV, respectively, as shown in Figs. 10 and 11. Accordingly, the corresponding J_d of the pixels of the $In_{0.53}Ga_{0.47}As$ and the $In_{0.75}Ga_{0.25}As$ FPAs also scattered over relatively large ranges, respectively. Considering the fact that the temperature-dependent IV measurements were carried out for merely a randomly selected pixel for each FPA, a random deviation from the average value of the measured dark current is inevitable, even though the pixel selected was near the center region of the FPA chip. Second, another point that must be noted is that the FPAs were operated in nominal zero-bias conditions. However, the practical offset voltage in the input-stage of the CITA typically ranges from -5 to 5 mV. This factor also contributes the discrepancy of the dark current density to some extent.

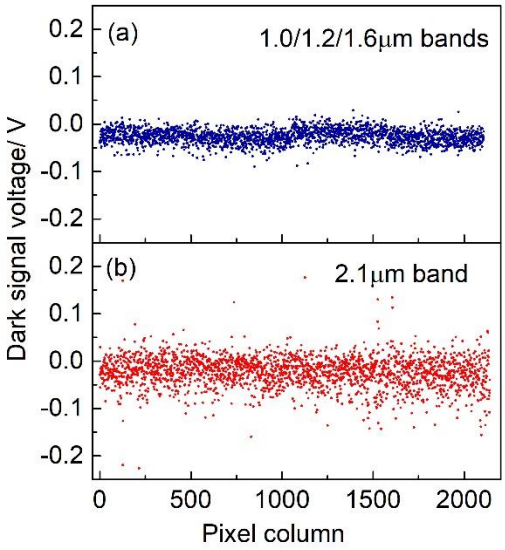


Fig. 10. Measured dark signal voltage curves of a row of pixels for FPA assemblies (a) A at 293 K and high gain and (b) B at 223 K and low gain.

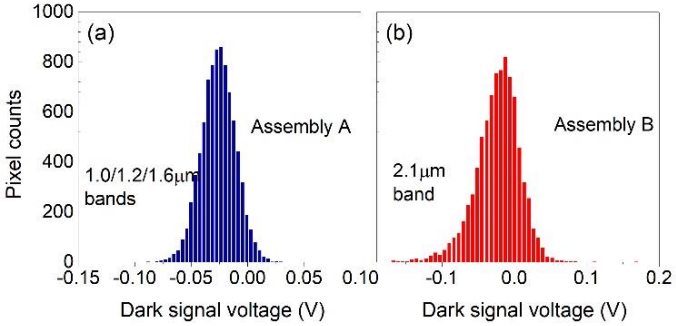


Fig. 11. Statistical histogram distributions of the measured dark signal voltages for FPA assemblies (a) A at 300 K and high gain and (b) B at 223 K low gain.

C. Detector readout noises

Dark noise measurements were performed in the same system under the same operation parameters. The total dark noise voltage of the pixel (i, j), V_{dn}(i, j), was taken as the standard deviation of the output dark signal.

$$V_{dn}(\mathrm{i,j}) = \frac{1}{K}\sqrt{\frac{1}{F-1}\{V_{dn}[(\mathrm{i,j}),\mathrm{f}] - \overline{V_{dn}}[(\mathrm{i,j})]\}^2}$$

where K is the amplifier gain factor of the measurement system, F is the total grabbed frame number, $V_{dn}[(i,j),f]$ is the dark noise of pixel (i, j) at the f frame, $\overline{V_{dn}}[(\mathrm{i,j})]$, is the mean dark noise of pixel (i, j) averaged over F frames. In order to ensure a high degree of accuracy of the noise data, 100 frames were acquired for each noise measurements. Figs. 12(a) and 12(b) plot the measured V_{dn} of pixels for both assemblies. The calculated mean V_{dn} over the whole array ($\overline{V_{dn}}$) are 1.8 and 1.1 mV for FPA assemblies A and assembly B, respectively. Such readout noise levels well satisfied the mission requirements to enable the SNR of higher than 400, 250,180 and 100 for the 1.0, 1.2, 1.6 and 2.1 m channels, respectively. Statistical histograms revealed asymmetrical Gaussian distributions for the two FPA

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assemblies, as shown in Figs. 13(a) and 13(b), respectively. By comparing Figs. 13 and 11, it is suggested that a portion of the pixels behave as higher noisy though their dark current remain relative low. Such a phenomena is more pronounced for the extended-wavelength $In_{0.75}Ga_{0.25}As$ FPA which is associated with the defect-related tunneling of dark carriers as this process is proportional to the lattice-mismatch degree.

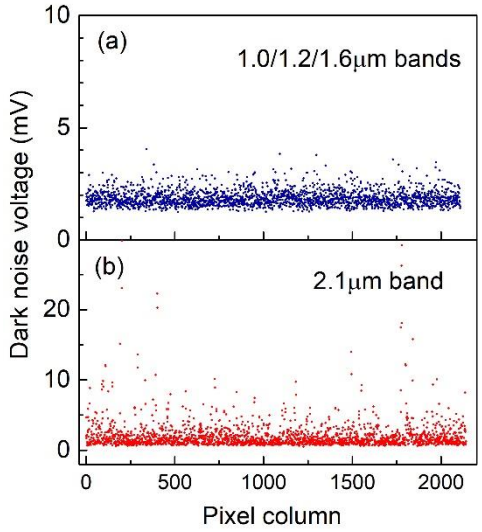


Fig. 12. Measured dark noise voltage curves of a row of pixels for FPA assemblies (a) A at 300 K and high gain and (b) B at 223 K low gain.

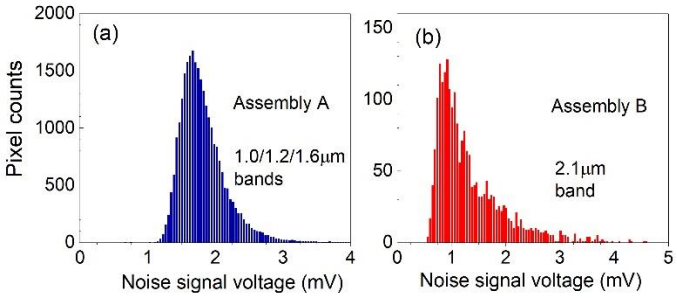


Fig. 13. Statistical histogram distributions for the measured dark noise voltages for FPA assemblies (a) A at 300 K and high gain and (b) B at 223 K low gain.

D. Modulation transfer function

MTF provides a measure of image resolution or spatial frequency response of the infrared imaging system or component. It reflects how the contrast is transferred from object space to image as a function of spatial frequency. Figure 14 shows the MTF measurement results of the FPA assemblies carried out by using a self-build system based on a slit scan method with an F/2 optics. The curves were deconvoluted from the measured raw data to subtract the MTF of the optics system. The horizontal and vertical MTFs at Nyquist frequency based on the pixel pitch of 30 μm are about 0.504 and 0.542 for assembly A and B, respectively, at 16.67 cycles/mm. By comparing, such a result clearly indicates that the mesa isolation structure of the 2.1 μm $In_{0.75}Ga_{0.25}As$ FPA chip benefits a better MTF by comparing with the planar $In_{0.53}Ga_{0.47}As$ chip for the 1.0, 1.2 and 1.6 μm channels.

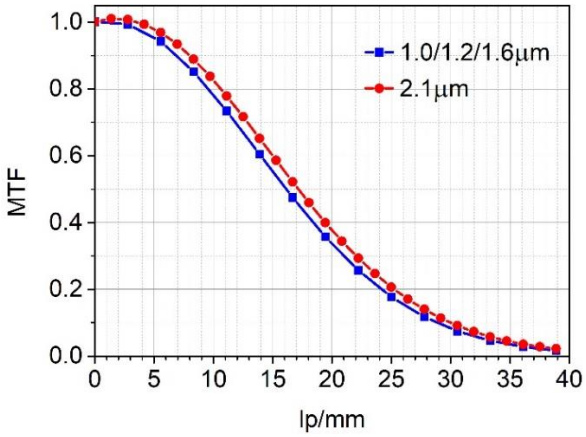


Fig. 14. Measured MTF for the two FPA assemblies by using a slit method.

E. Performance comparison and imaging demonstration

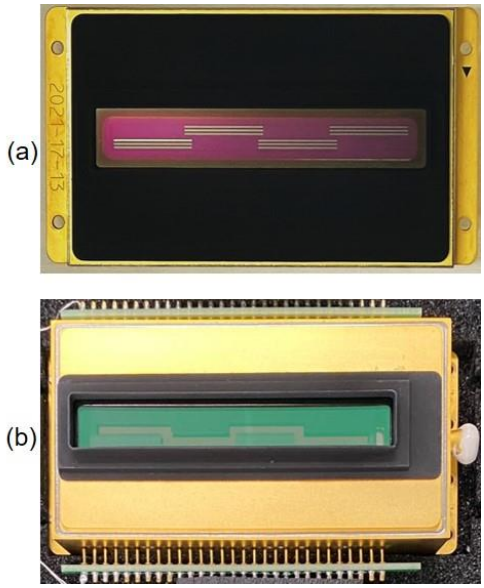


Fig. 15 Photographs of the finished flight-level FPA detector assemblies (a) A and (b) B.

Figures 15(a) and 15(b) show photographs of the finished FPA assemblies A and B, respectively. A Lyot aperture was designed for each assembly to suppress the spray lights inside the PMRSI system. The FPA assemblies have passed the screen, the acceptance, and the qualification levels reliability tests that contain thermodynamics, mechanics, and radiation. Complete performance parameters for the two FPA assemblies are listed in table I to give a more direct comparison. The flight-level have achieved pixel operabilities of 99.99% and 99.78% for the assemblies A and B, respectively. Very small response non-linearities of 0.7% and 2.5% were measured for well occupations between 10% to 90%. Noteworthy is that the power consumption in Table I refers to only the ROIC power. Since the TEC power consumption of the detector assembly B closely depends on the thermal design and cooling power of the

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TABLE I A LIST OF THE PERFORMANCE PARAMETERS FOR THE TWO FPA DETECTOR ASSEMBLIES MEASURED AT A TINT OF 14MS UNDER ITR MODE.

Spectral band /μm	**1.0**	**1.2**	**1.6**	**2.1**
FPA size		2048×52		2048×12
Channel array size	2048×4	2048×4	2048×4	2048×4
Quantum efficiency / %	81	82	79	82
Average dark signal / mV		16.8@HG		22.6@LG
Dark noise / mV		1.8@HG		1.1@LG
Dark current density/ nA/cm^2		48@293K		320@223K
Pixel operability / %		99.99@HG		99.78@LG
Response non-linearity / %		0.7		2.5
Response non-uniformity / %	3.0	3.0	1.9	3.2
Full-well capacity / e$^-$		300 K@HG, 1.6 M@LG		
Max. readout rate / MHz		10		
Readout mode		IWR / ITR / Lineselection		
Peak detectivity / cmHz$^{1/2}$/W	3.3×10^{12}	3.9×10^{12}	4.6×10^{12}	1.4×10^{12}
MTF		0.50		0.54
FPA chip temperature / K	293	293	293	223
Power consumption/mW		320		110



Fig. 16. A scanning spectral imaging demonstration under daylight background for channels of (a) 1.6 and (b) 2.1 μm acquired at gain settings of high and low for assemblies A and B, respectively, and at the same integration time of 7.5 ms.

instrument, here only the ROIC power is listed. The actual TEC power for ground test is about 44 W (12V/3.7A) for the FPA temperature of 223 K. A liquid-cycling cooling system with a mounting stage temperature of 278 K is utilized for the heat sink.

Ground scanning imaging demonstrations during a bright daylight background by using one of the five SWIR camera modules of the PMRSI system were performed to evaluate the imaging capability. Operating temperatures for the FPA detector assemblies A and B were 293 and 223 K, respectively.

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TABLE II A COMPARISON OF THE PERFORMANCE PARAMETERS OF INGAAS FPA SENSORS DEPLOYED IN REMOTE SENSING INSTRUMENTS FOR AEROSPACE APPLICATIONS.

Satellite	Instrument	Country	Launch year	Swath width /km	Ground resolution /m	FPA detector assembly performances				
						Bands /μm	Pixel array size	Pitch /μm	Material type	Pixel operability / %
HY-3A	PMRSI	China	2023	1000	100	2.1	2048×12	30×30	$In_{0.75}Ga_{0.25}As$	99.8
						1.0/1.2/1.6	2048×52	30×30	$In_{0.53}Ga_{0.47}As$	100
FY-3G	SWIR PMAI	China	2022	2900	500	1.38/1.64	320×256	25×25	$In_{0.53}Ga_{0.47}As$	100
FY-4B	GHSI	China	2020	2000	0.5	1.38/1.64	1024×2	20×20	$In_{0.53}Ga_{0.47}As$	100
PACE	OCI	USA	2024	2500	1080	1.0/1.2/ 1.6/2.1	8×2	200×20 0	$In_{0.53}Ga_{0.47}As$	NG
Proba-V	Vegetation-P	EU	2013	2285	300	0.9-1.7	3000×1	25×25	$In_{0.53}Ga_{0.47}As$	99.6
EarthCARE	MSI	EU	2014	150	500	1.55/2.1	512×1	30×30	$In_{0.53}Ga_{0.47}As$ $In_{0.75}Ga_{0.25}As$	≥99.0
LCROSS	Near Infrared Camera	USA	2009	100	100	0.9-1.7	320×256	30×30	$In_{0.53}Ga_{0.47}As$	≥99.0
Envisat	SCIAMACH Y	EU	2002	1150	1000	1-1.75 /2.0/2.3	1024×1	25×500	$In_{0.53}Ga_{0.47}As$ $In_{0.603}Ga_{0.40}As$ $In_{0.81}Ga_{0.19}As$ $In_{0.83}Ga_{0.17}As$	≥99.0

Figures 16(a) and 16(b) give a spectral imaging demonstration for channels of 1.6 and 2.1 μm acquired at gain settings of high and low, respectively, and a T_{int} of 7.5 ms. The images are raw without any software correction. While very high contrasts are observed for both images, a closer look at the image details reveals apparent different grayscales for the glasses of windows on the same building. Darker glasses are seen for the 1.6 μm channel whereas the glasses tend to be light gray in the 2.1 μm channel. This suggests that the extended spectral range towards longer wavelength in the SWIR region has an added benefit of recording heat signatures without sacrificing its reflective optical capabilities. Meanwhile, foliage penetration imaging capabilities can also clearly seen for both channels, which proves the advantage of the imaging features of the SWIR bands.

Table II summarizes a comparison of the performance parameters of InGaAs FPAs that were deployed in primary aerospace instruments over the world. It is seen that both the 2048 × 52 $In_{0.53}a_{0.47}As$ and 2048 × 12 $In_{0.75}Ga_{0.25}As$ FPA detectors developed for the PMRSI in HY3A achieved the highest pixel operabilities. Meanwhile, the array size of the extended-wavelength InGaAs FPA here is also record large for aerospace sensing application by far. Furher development challenges of the InGaAs FPAs rely on the continually improvement of the material quality and the better pn junction formation as well as surface passivation technologies as for high-end SWIR astronomy observing and some geo-stationary orbit satellites extremely low readout noises and very long exposure time at high gain are generally required.

IV. CONCLUSION

In conclusion, the design, fabrication and characterization of $In_{0.53}Ga_{0.47}As/InP$ and $In_{0.75}Ga_{0.25}As/In_{0.52}Al_{0.48}As$ FPA detector assemblies developed for the PMRSI on the HY3A satellite are reported. Large pixel array in formats of 2048×52 and 2048×12 are realized for the detector assemblies A and B, respectively, among which the array length of the extended-wavelength $In_{0.75}Ga_{0.25}As$ FPA is as far as know the longest for aerospace applications. Four densely arranged wavelength bands with center wavelengths of 1.020, 1.245, 1.640 and 2.135 μm, respectively, are monolithically achieved by integrating of optical filters. Low noise ROICs by incorporating of a tri-S/H architecture are designed. Space-qualified FPA detector assemblies with low dark current densities of 48 and 320 nA/cm^2, high quantum efficiencies of >80%, low readout noises of 1.8 and 1.1 mV, high pixel operabilities of 99.99% and 99.78%, good response nonuniformities of 3.0%, high MTFs of 0.5, and high peak detectives of 4.0×10^{12} and 1.4×10^{12} $cmHz^{1/2}/W$ are simultaneously obtained, respectively. Laboratory imaging demonstrations highlights the superior performance for different channels. High quality spectral imaging data for the ocean science have already been continuously produced since 2025 after in-orbit calibration.

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